Team Eta

Project Viva

EE 322: Analog and Mixed Signal Circuits

1. INTRODUCTION

An analog synthesiser is a synthesiser that uses analog circuits and analog signals to generate sound electronically[1]. Traditional analog synthesisers are often complex, expensive, and difficult for beginners to understand.

> Music synthesisers are used to create sounds electronically by generating and modifying waveforms, and are a foundational instrument for electronic music.
>
> This project aims to make a functional, compact, and easily understandable analog music synthesiser using fundamental electronic components integrated on a PCB. It offers a hands-on platform for learning about signal generation, processing, and amplification.

2. PROBLEM STATEMENT

Most commercially available analog synthesisers are expensive and rely on complex circuit architectures, making them inaccessible to beginners. The problem addressed in this project is the lack of an easily understandable, low-cost analog synthesiser circuit that demonstrates core concepts like waveform generation and sound processing methods.

Problem Statement: To design and build an analog music synthesiser that will generate musical tones corresponding to the C major scale using oscillators, process and combine the waveforms to produce polyphonic sound, and deliver an amplified output through a speaker.

Developing a simplified analog synthesiser will aid in practical learning and serve as a tool to visualise and hear how resistors, capacitors, and operational amplifiers (op-amps) shape sound in real time.

3. PROPOSED APPROACH

a) Overview of Approach

The synthesiser will be constructed from modular analog building blocks. A signal generator is designed using 8 oscillators (astable multivibrators) based on operational amplifiers, each tuned to a specific musical note in the C major scale (C4–C5). The frequency corresponding to each note is selected according to the figure below (Fig. 1), for octave 4.

Note	Octave 0	Octave 1	Octave 2	Octave 3	Octave 4	Octave 5	Octave 6	Octave 7	Octave 8
C	16.35 Hz	32.70 Hz	65.41 Hz	130.81 Hz	261.63 Hz	523.25 Hz	1046.50 Hz	2093.00 Hz	4186.01 Hz
C#/Db	17.32 Hz	34.65 Hz	69.30 Hz	138.59 Hz	277.18 Hz	554.37 Hz	1108.73 Hz	2217.46 Hz	4434.92 Hz
D	18.35 Hz	36.71 Hz	73.42 Hz	146.83 Hz	293.66 Hz	587.33 Hz	1174.66 Hz	2349.32 Hz	4698.63 Hz
D#/Eb	19.45 Hz	38.89 Hz	77.78 Hz	155.56 Hz	311.13 Hz	622.25 Hz	1244.51 Hz	2489.02 Hz	4978.03 Hz
E	20.60 Hz	41.20 Hz	82.41 Hz	164.81 Hz	329.63 Hz	659.25 Hz	1318.51 Hz	2637.02 Hz	5274.04 Hz
F	21.83 Hz	43.65 Hz	87.31 Hz	174.61 Hz	349.23 Hz	698.46 Hz	1396.91 Hz	2793.83 Hz	5587.65 Hz
F#/Gb	23.12 Hz	46.25 Hz	92.50 Hz	185 Hz	369.99 Hz	739.99 Hz	1479.98 Hz	2959.96 Hz	5919.91 Hz
G	24.50 Hz	49 Hz	98 Hz	196 Hz	392 Hz	783.99 Hz	1567.98 Hz	3135.96 Hz	6271.93 Hz
G#/Ab	25.96 Hz	51.91 Hz	103.83 Hz	207.65 Hz	415.30 Hz	830.61 Hz	1661.22 Hz	3322.44 Hz	6644.88 Hz
A	27.50 Hz	55 Hz	110 Hz	220 Hz	440 Hz	880 Hz	1760 Hz	3520 Hz	7040 Hz
A#/Bb	29.14 Hz	58.27 Hz	116.54 Hz	233.08 Hz	466.16 Hz	932.33 Hz	1864.66 Hz	3729.31 Hz	7458.62 Hz
B	30.87 Hz	61.74 Hz	123.47 Hz	246.94 Hz	493.88 Hz	987.77 Hz	1975.53 Hz	3951.07 Hz	7902.13 Hz

Figure 1. Music notes to frequencies chart (Image Source: [2])

Each note is gen by a single osc, played using a push button. Then there’s a buffer.

All 8 oscillator outputs will feed into an analog summing amplifier (adder) that combines these signals, allowing us to simultaneously press multiple keys.

The resulting waveform will then be passed through a final audio amplifier stage

Amplified output will be passed thru a buffer and then delivered through a speaker.

Power will be supplied through a dual ±8 V regulated source, and the entire system will be designed for PCB integration.

b) Components and Blocks

The outline of the circuit is given below (Fig. 2).

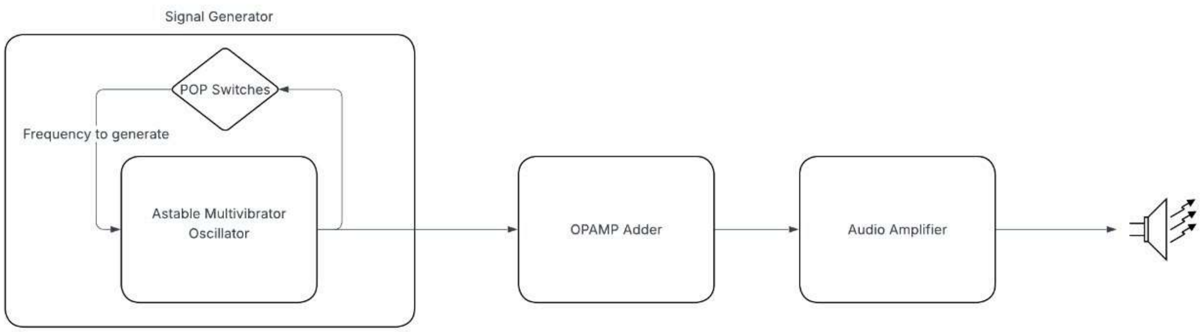


Figure 2. System-Level Block Diagram

1. Signal Generator:

   The signal generator is realised using an **astable multivibrator circuit**, which doesn't require a dedicated input and generates square waves of the required frequency, which are then further processed.

   Q's

   a. Why astable not sine or 555 timer - ring oscillators = high Rs and Cs. initially square not sine bcoz we wanted to process it, it is easy to get sine,triangle from square by low pass and integrate. Most common examples we found was astable multivibrator. And 555 timer, we were not sure about it because it had digital component.Why tl084. This because it has high slew rate 20 and 741 has 0.7 V/us. slew rate{max rate at which output voltage can change over time} and it is good for audio signal as it will help change it fast without distortion. TL084 and TL074 are identical, just TL084 was available

   b. Component values (R, C). We used the formula 1/2RC(1+2R1/R2). ins

Editor - C:\Users\anujc\OneDrive\Documents\MATLAB\Analogproject.m

main.m | SSRP.m | Analogproject.m

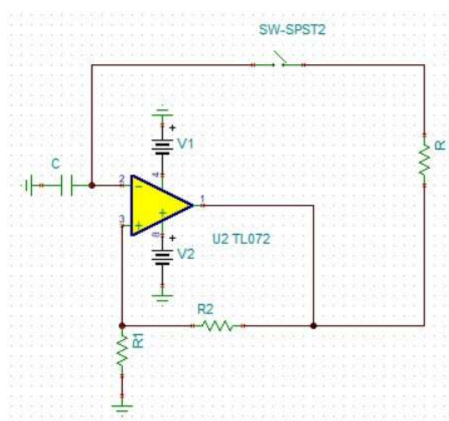
```
R1=5.6e3;
R2=5.6e3;

lambda=R1/(R1+R2);

C=220e-9;

f=[261.63 293.66 329.63 349.23 392 440 493.88 523.25]; %list of frequencies

R=1./(f.*2.*C.*log((1+lambda)/(1-lambda)))
```

Command Window

New to MATLAB? See resources for Getting Started.

```
>> Analogproject

R =

   1.0e+04 *

    8.1827    7.2902    6.4947    6.1302    5.4613    4.8656    4.3348    4.0914

>> Analogproject

R =

   1.0e+03 *

    7.9071    7.0446    6.2759    5.9237    5.2774    4.7016    4.1887    3.9536
```

R1 << R2/2 for R to be in the range of 50 to 100k. Because of more precision and exact R we took the value of R according to the R available in the lab.

c. 

2. Adder:

   We added buffer before Adder as we wanted to isolate the oscillator part from the adder as the 1k resistor we used for adder might interfere with the resistors we are using in the oscillator.

The adder is non inverting adder. We just used values 1k and 1ks.

To enable us to play multiple notes simultaneously, each oscillator's output is fed through an adder circuit made using an op-amp. It linearly combines the active oscillator outputs to allow for multiple notes (polyphony), then feeds the composite waveform to the amplifier. This will be overlooked by Himani.

3. Amplifier/TDA buffer:

   We used **TDA 2030** IC which is typically used for AB class amplifier. Were planning to use the circuit which was given in the datasheet. LOW freq class AB amplifier.

   Class AB has good linearity and efficiency between A and B. A has high higher linearity and low efficiency and B has better efficiency and very low linearity.

   Amplifiers are classified based on very good linearity less efficiency to very high efficiency and low linearity.

   We realised we don't need amplifier as we already have high voltage signal. We needed current to drive the speaker. We used TDA 2030 as a buffer which can drive the 8ohm speaker.
   Q. Why tda 2030 as buffer not any opamp>
   BEcause tda 2030 has output current of 3.5A. For us to run an 16V ptp signal with 8ohm speaker we needed 2A. This fits. TL072 has output current in mAs.

   This stage takes the output of the adder (summed waveform) and amplifies it as needed, and the amplified wave is played through a speaker. This part of the circuit will be overlooked by 2 members (Himani and Ineesha)

c) Advantages of the Proposed Approach:

low-cost alternative to traditional commercial designs

simplifying its function into a few fundamental blocks.

As it uses basic components such as resistors, capacitors, and op-amps, we can visually and audibly understand fundamental analog concepts like oscillation, summing, and amplification.

Its modular structure with distinct oscillator, adder, and amplifier stages simplifies analysis and debugging, while the polyphonic capability allows multiple notes to be played simultaneously. SO TRUE THO

4. WORKING OF THE CIRCUIT

a) Signal Generator

Each note (selected using pushkeys) is generated using an astable multivibrator circuit, as shown in Fig. 3 below.

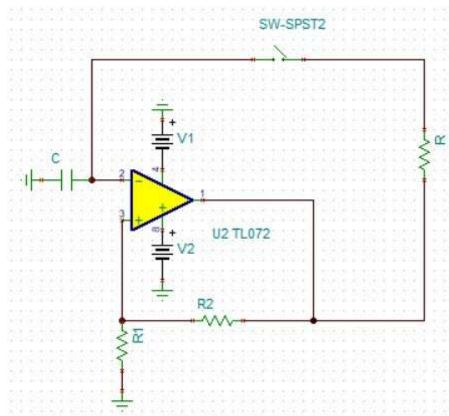


Figure 3. Circuit diagram of an Astable Multivibrator

The values of R, C and R_1 and R_2 are chosen such that the output is a square wave of the desired frequency (f_0) of the note to be played. The following formula is used:

$$f_0 = \frac{1}{T} = \frac{1}{2RC\ln(1+2R_1/R_2)}$$

Figure 4. Formula used to calculate the values of resistance and capacitance for a given frequency (Source: [3])

b) Adder Circuit

The square waves generated by the oscillators are passed through the adder made using an op-amp (TL072). The output would be the sum of each signal passed in, such that when we press multiple keys, each of the signals of corresponding frequencies is added and present in the output.

c) Amplifier

This stage amplifies the output of the adder (the summed waveform) and plays it through a speaker.

5. REFERENCES

1. “Analog synthesizer,” Wikipedia, The Free Encyclopedia. [Online]. Available: https://en.wikipedia.org/wiki/Analog_synthesizer
2. MixButton, “Music note to frequency chart,” MixButton.com, [Online]. Available: https://mixbutton.com/music-tools/frequency-and-pitch/music-note-to-frequency-chart
3. S. Franco, Design with Operational Amplifiers and Analog Integrated Circuits, 4th ed. New York, NY, USA: McGraw-Hill Education, 2015.

4.